TITLE PAGE

- Problem Statement ID: FT004
- Problem Statement Title:Invoice Financing for MSMEs
- Domain: Fintech
- Team Name: Mindstack
- Team Leader Name: Kartik K. Shivankar

IDEA TITLE- PayNidhi : AI-Powered Invoice Liquidity Platform

Proposed Solution

Proposed Solution (For FT004 Invoice Financing):

Small businesses (MSMEs) in India face severe cash flow crunches due to delayed payments (30-90 days), while lenders fear "fake invoices" and defaults. Our platform bridges this trust gap using a secure, tech-driven financing engine compliant with the Factoring Regulation Act, 2011.

Following is our Idea/Approach:

- **Multimodal AI-Extraction:** Instantly extract IRN, GSTIN, and financial totals from uploaded PDFs/images.
- **GST-Trust Verification:** Cross-references the Invoice Reference Number (IRN) against a mock government registry for authenticity.
- **Double-Sells Prevention:** Utilizes Cryptographic Hashing to ensure a single invoice isn't financed on multiple platforms.
- **Three-Way Reconciliation:** AI-driven matching of Invoices with Purchase Orders (PO) and Goods Received Notes (GRN).
- **Digital Acceptance Portal:** A secure "Handshake" interface for Buyers to legally approve and assign debt to Lenders.
- **Automated Liquidity Engine:** A backend Escrow Splitter that

User Flow Diagram

Uniqueness

Smart Escrow Splitting: Automated "Splitter" logic ensures the Lender is paid first from the Virtual Account, eliminating the risk of the Seller diverting funds.

3-Pillar Security: Combines Virtual Accounts, a 20% "Haircut" (margin), and automated e-NACH chargebacks to ensure the Lender's capital is always protected.

Real-Time GST Verification: Uses AI-driven OCR to instantly validate Invoice Reference Numbers (IRN) against government data, preventing fraud and "fake" invoices.

How It Addresses the Problem

Liquidity Crunch for MSMEs?

We convert unpaid invoices into instant cash flow, providing 80% of the value upfront so businesses can reinvest without waiting for credit cycles.

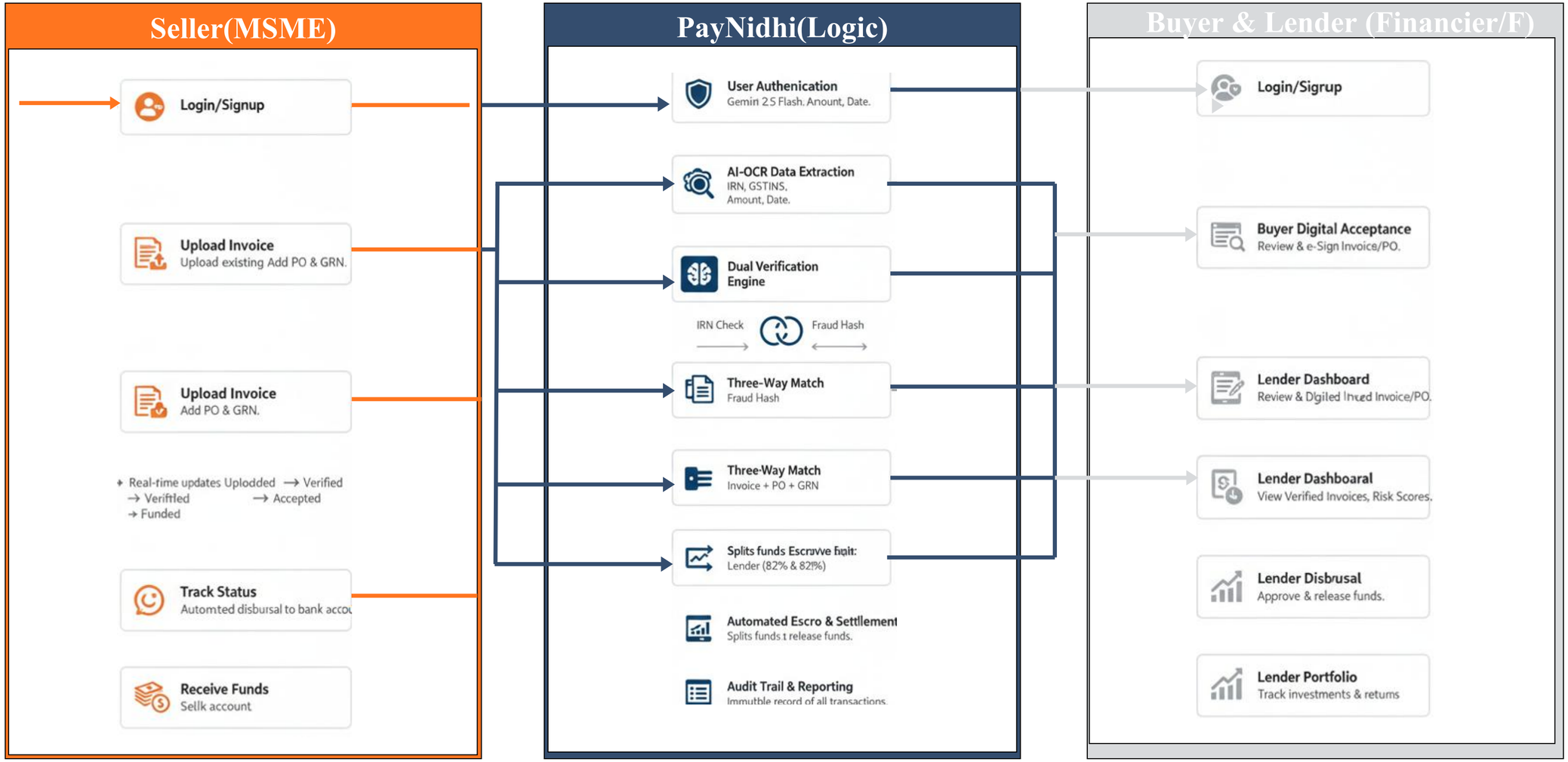
Lender Trust and Payment Diversion?

Our automated Escrow-Split system and GST verification ensure that the invoice is real and the lende gets paid first, removing the risk of "fake" debt or stolen payments.Your paragraph text

Buyer Onboarding Resistance?

Our Recourse model eliminates financial liability for the Buyer. By removing legal hurdles, we ensure

Process flow diagram



Methodology used

The PayNidhi Operational Procedure

A Seamless 5-Step Journey from Invoice to Cash

1. Intelligent Ingestion

The Seller uploads their existing invoice (PDF/Image) along with supporting documents (PO/GRN). The system uses Contextual AI to instantly digitize the data, ensuring that the MSME doesn't have to change their current billing habits.

2. The Trust Validation

Before the invoice is visible to any lender, it passes through a Double-Gate Security Check:

- **Government Alignment:** The IRN is verified for authenticity.
- **Integrity Lock:** A unique digital fingerprint is created to ensure this specific invoice hasn't been financed elsewhere (Anti-Fraud).

3. Collaborative Confirmation

The platform notifies the Buyer to review the digitized invoice. By clicking "Accept," the Buyer confirms the debt is valid and the goods were received. This "Digital Handshake" converts a piece of paper into a legally backed financial asset.

4. Real-time Liquidity Auction

Once verified and accepted, the invoice appears on the Lender's Dashboard. Lenders can see the "Trust Score" and the verified status, allowing them to approve the funding with a single click.

5. Automated Settlement & Split

- **Neutral Escrow:** Buyers pay into a unique Virtual Account, ensuring the platform (not the seller) controls the repayment flow to protect the lender.
- **Smart Split Logic:** The system instantly divides the incoming funds—routing the Principal + Interest to the Lender and the Retention Balance to the MSME.

Solution Concept & Feasibility

Operational Feasibility

- **System Integration:** Uses RESTful APIs to ensure compatibility across web platforms and mobile devices for Sellers and Lenders.
- **User-Centric Interface:** Features a Progressive Dashboard that live-updates the status of invoices from "Uploaded" to "Funded," making the complex credit cycle transparent.
- **Data Security & Privacy:** Guarantees absolute confidentiality for sensitive financial documents by utilizing End-to-End Encryption and secure AWS/Google Cloud storage buckets.

Technical feasibility

- **AI-Driven Extraction:** Employs Gemini 2.5 Flash to provide near-instant, human-level accuracy in digitizing non-standardized invoice formats.
- **Fraud Prevention:** Implements a Dual-Layer Verification Engine (IRN Cross-Check + SHA-256 Hashing) to eliminate double-financing risks.
- **Scalable Architecture:** Built on the MERN Stack, allowing the platform to handle high volumes of concurrent transactions and real-time ledger updates.

Viability

- Risk Reduction: Enhances lender confidence by reducing fraud probability by over 90% through automated three-way matching.
- Economic Impact: Facilitates MSME cash flow, potentially bridging the credit gap by providing liquidity in under 24 hours compared to the traditional 30-day wait.
- Cost Efficiency: Uses open-source MERN libraries and pay-per-use AI models to keep operational cost low and system flexibility high.

Challenges

- **High Fraud Risks:** Sellers might upload the same invoice to multiple platforms.
- **Manual Data Entry Errors:** MSMEs struggle with complex digital forms.
- **Complex Repayment:** Ensuring the lender gets paid back on time by the buyer.

Solution

- **Unique Hashing:** Generate a cryptographic fingerprint for every invoice to block duplicate entries.
- **AI-OCR Ingestion:** Use Gemini to automatically extract data directly from existing invoice photos or PDFs.
- **Virtual Escrow:** Assign unique virtual accounts to each invoice so payments are captured and split automatically.

Feasibility through Digital Transformation

- **Paperless Credit:** Converts physical, stagnant invoices into digital, liquid financial assets.
- **Democratizing Finance:** Enables small rural MSMEs to access institutional credit based on Invoice Strength rather than just collateral.
- **Aligns with 'Digital India':** Directly supports GST initiatives and digital payment adoption for small business owners.

Use cases & description

Primary Usecase

Automated MSME Credit Access:

Enables Micro, Small, and Medium Enterprises (MSMEs) to convert their unpaid invoices into immediate working capital by connecting them with institutional lenders through a secure digital escrow bridge.

Workflow

AI Extraction: Tesseract instantly pull IRN and financial data from uploaded invoices.

Fraud Validation: Validation Engine cross-checks GST records to block fake or double-financed invoices.

Buyer Acceptance: Corporate buyers digitally sign to legally acknowledge the debt in the Smart Ledger.

Escrow Split: System triggers automated 80/20 disbursement via Virtual Accounts to the Seller and Lender.

Target User	Application Scenario	Key Benefit
Manufacturing MSMEs	Bridging the credit gap for raw material procurement during peak production cycles.	Improved cash flow velocity by 85% without traditional collateral.
Institutional Lenders	Accessing a pre-verified, AI-audited pool of MSME invoices for low-risk, high-frequency short-term lending.	Automated risk assessment powered by the **Validation Engine (Mistral + LangChain)**.
Corporate Buyers	Streamlining supply chain stability by ensuring vendors are paid instantly, reducing delivery delays.	Enhanced supplier relations and digitized accounts payable via the **Smart Ledger**.
Rural Supply Chains	Utilizing **Offline-First PWA (IndexedDB)** to upload and manage invoices in areas with low internet connectivity.	Inclusive financial access for underserved regional businesses.

Technology stack used

Tech stack

NEXT.js React — backend and frontend

aws ONNX — Cloud Infrastructure

stripe — Strip API

Firebase

Google Colab

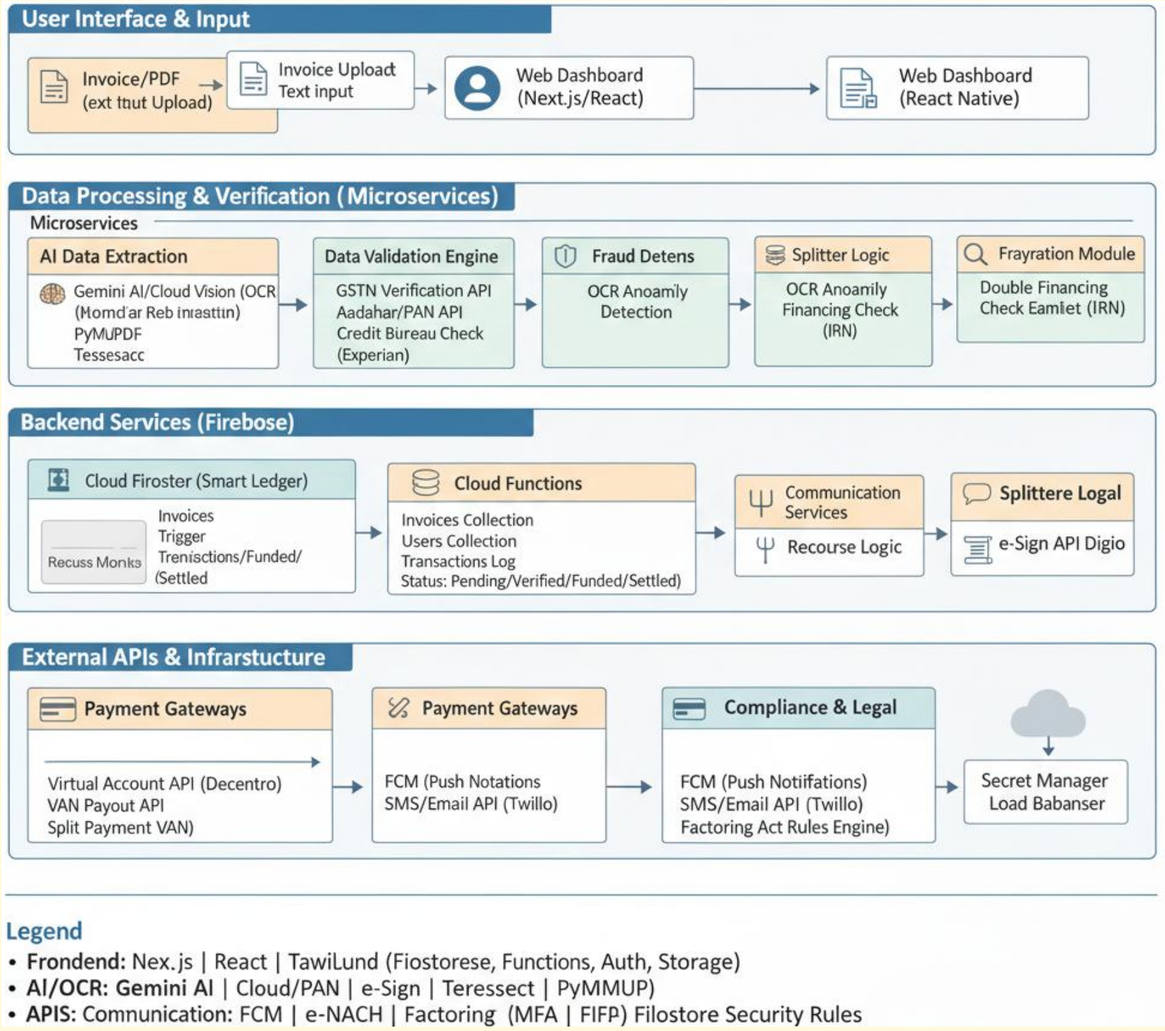
Gemini AI

Tesseract — OCR Model

Twilio

- **Next.js | React | Tailwind CSS:** Fast, responsive frontend for Seller, Buyer, and Lender dashboards.
- **Firebase Auth & Cloud Storage:** Secure user login (SSO) and encrypted storage for sensitive PDF invoices.
- **Cloud Firestore:** Real-time NoSQL database to track live invoice statuses (Verified $\rightarrow$ Funded $\rightarrow$ Settled).
- **Gemini AI & Vision API:** Intelligent OCR to extract IRN and verify invoice data instantly.
- **Firebase Cloud Functions:** Serverless "Splitter" logic to automate fund distribution and platform fees.
- **FCM & e-NACH Integration:** Real-time push alerts for funding and automated "Recourse" recovery.

Architecture diagram



Constraints

Potential Challenges & Risks	Strategic Solutions
• **Double Financing Fraud:** The high risk of a Seller pledging a single Invoice Reference Number (IRN) to multiple lenders or platforms simultaneously. • **Onboarding Friction:** High resistance from large Corporate Buyers to integrate their internal ERPs or procurement workflows with a third-party platform. • **Collusive Circular Trading:** The danger of "shell companies" (Seller and Buyer) creating fake trade transactions to siphon credit from Lenders. • **Credit Rating Volatility:** Rapid downgrades in a	• **Escrow-Based Disbursement:** Routing all buyer payments through unique Virtual Accounts (VAN) to ensure the "Splitter" script handles repayments automatically. • **e-NACH Digital Mandates:** Securing a legally binding standing instruction at the time of onboarding to enable automated "Chargebacks" from the Seller. • **Resolution Holdback:** Automatically pausing the final 20% disbursement if a "Dispute" flag is raised, protecting the Lender's principal. • **Dynamic Margin Adjustment:** Adjusting the "Haircut"